Design and Implementation of an R/C/Open Identification and Capacitance Measurement System

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1 Introduction

In this project, a measurement system was designed and implemented to identify the object connected to the measurement terminals as a capacitor, resistor, or open circuit. When a capacitor is connected, the system calculates and displays its capacitance. However, the target measurement range is limited to resistors from 1 kΩ to 10 kΩ and capacitors from 1 nF to 10 nF. Within the target range, capacitance measurements can be performed with an accuracy of 0.97% to 3.5%.

The proposed system uses different measurement methods depending on the connected object. For capacitance measurement, the microcontroller measures the periodic signal generated by the RC charging/discharging circuit and the comparator. The capacitance is then calculated from the measured period using a conversion equation. For an open-circuit and a resistor classification, the output voltage of the comparator is measured by the microcontroller and compared to the predefined threshold voltage for classification. The identification result and measured value are shown on the display.

2 System Overview and Circuit Schematic

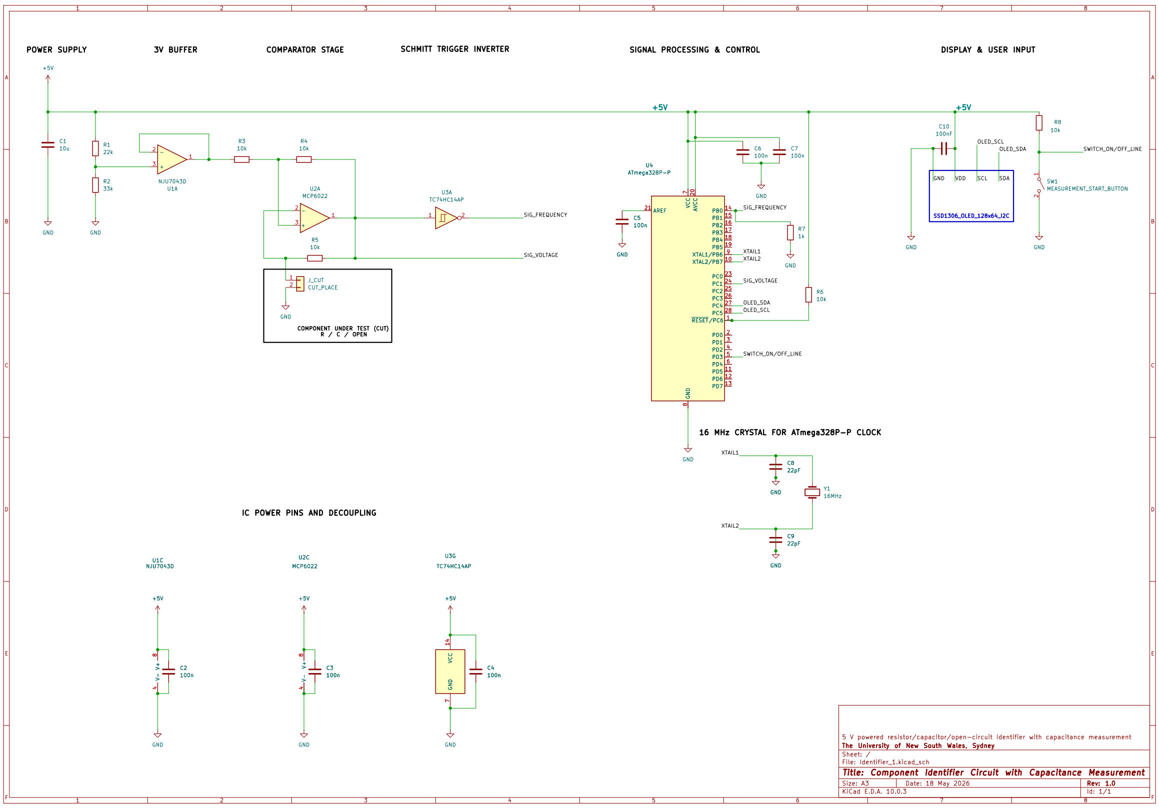

Figure 1: Circuit schematic

The system consists of a power supply stage, buffer, comparator, Schmitt trigger inverter, microcontroller, display, and user switch. A 5 V DC supply is used to power the circuit. A voltage divider and buffer provide 3 V to the comparator. The comparator output is fed to the ATmega328P as the voltage detection signal, `SIG_VOLTAGE`. The same output is also routed through a Schmitt trigger inverter and supplied to the ATmega328P as the frequency measurement signal, `SIG_FREQUENCY`.

After power is applied, the system remains in a standby state until switch SW1 is turned ON. Once SW1 is activated, the ATmega328P detects a LOW signal on `SWITCH_ON/OFF_LINE` and begins component identification and measurement. First, the microcontroller reads the `SIG_FREQUENCY` signal to determine whether the component connected to the CUT is a capacitor. If a capacitor is detected, its capacitance is calculated from the measured signal period. If the connected component is determined not to be a capacitor, the microcontroller then reads the voltage level of `SIG_VOLTAGE` and identifies the component as either a resistor or an open circuit according to the predefined threshold value. Finally, the identification result is displayed and retained until SW1 is activated again. For a capacitor, the measured capacitance and the Schmitt trigger output period are displayed. For a resistor or an open circuit, the comparator output voltage, `SIG_VOLTAGE`, is displayed together with the identification result.

3 Design Theory

First, the hysteresis characteristic of the Schmitt trigger inverter was investigated. In this circuit, the Schmitt trigger inverter is used to form the comparator output into well-defined High and Low logic levels and to improve noise immunity around the switching thresholds. Since the datasheet only specifies the threshold voltages for supply voltages of 4.5 V and 6.0 V, the upper and lower threshold voltages at 5 V were measured experimentally. The resulting voltage transfer characteristic is shown in Fig. 2.

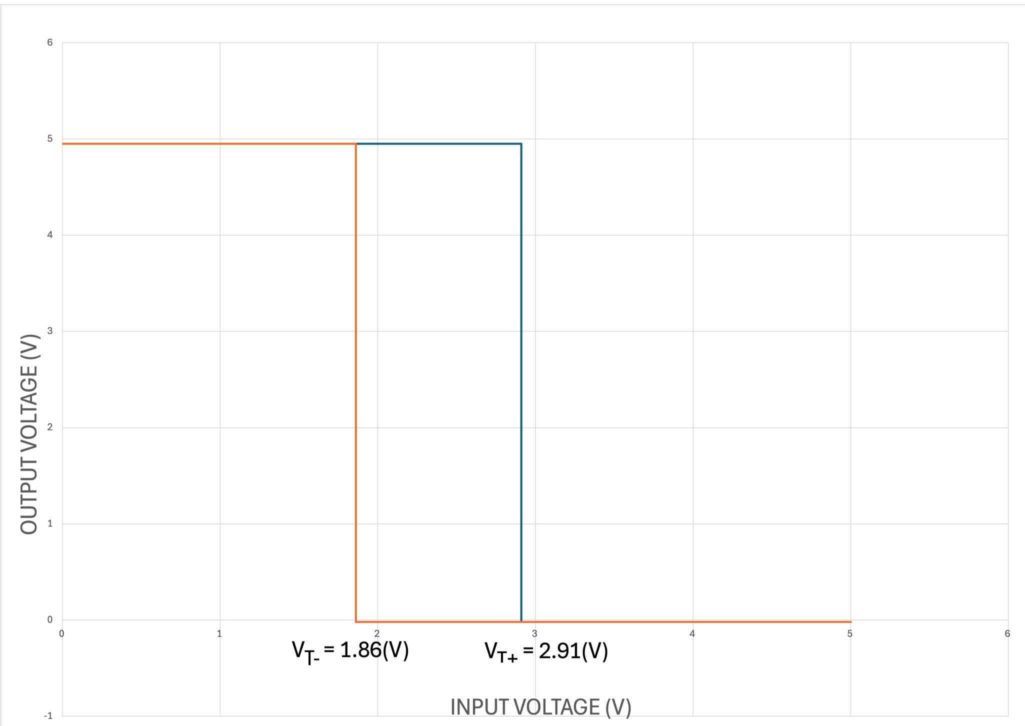

Figure 2: Voltage transfer characteristic of the 74HC14AP Schmitt trigger inverter at $V_{CC} = 5\,\mathrm{V}$.

3.1 Summary of Identification Method

Before discussing the identification method for each component, the overall identification process is summarized as follows.

1. SIG_FREQUENCY is checked to determine whether an oscillating signal is present. If oscillation is detected, the connected component is identified as a capacitor. Otherwise, the component is classified as either a resistor or an open circuit, and SIG_VOLTAGE is then measured.

2. SIG_VOLTAGE is used to distinguish between a resistor and an open circuit. As described in Sections 3.3 and 3.4, theoretically 3 V is measured when an open circuit is connected, whereas 5 V is theoretically measured when a resistor is connected. Therefore, by setting a threshold voltage between these two values, the ATmega328P can distinguish a resistor from an open circuit. In practice, the threshold voltage is determined experimentally.

For capacitance measurement, the frequency of SIG_FREQUENCY is measured. As described in Section 3.6, the measured period is substituted into the conversion equation to calculate the capacitance.

3.2 Capacitor Detection

First, the operation of the comparator when a capacitor is connected as the CUT is examined. The corresponding circuit is shown below.

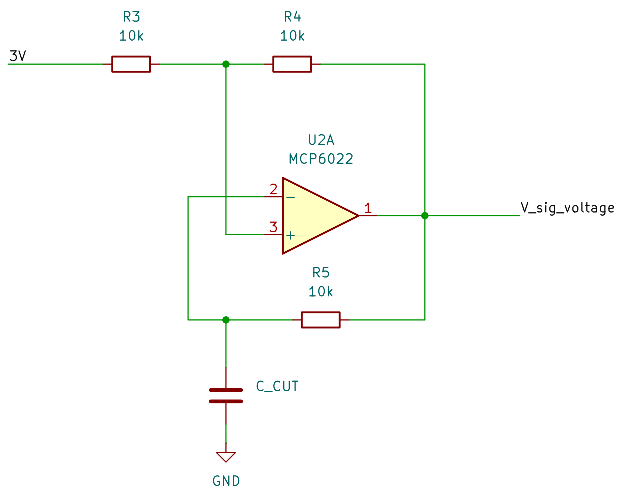

Figure 3: Comparator circuit when a capacitor is connected as the CUT.

Throughout this section, the output voltage labelled as V_sig_voltage in Fig. 3 is denoted by V_{sig}.

When a capacitor is connected as the CUT, it is assumed that both the inverting input voltage V_- and the output voltage V_{sig} are initially 0 V. Therefore, the initial voltage at the non-inverting input is

$$V_+ = \frac{3+0}{2} = 1.5 \text{ V}.$$

Since $V_+ > V_-$ and a positive feedback path is formed from V_{sig} through R_4 to V_+, the output voltage and non-inverting input voltage after a short settling period become

$$V_{\text{sig}} = 5 \text{ V}, \qquad V_+ = \frac{3+5}{2} = 4.0 \text{ V}$$

The capacitor then begins to charge through R_5, causing V_- to increase. When V_- exceeds $V_+ = 4.0$ V, the comparator output switches Low. Consequently,

$$V_{\text{sig}} = 0 \text{ V}, \qquad V_+ = \frac{3+0}{2} = 1.5 \text{ V}, \qquad V_- = 4.0 \text{ V} \tag{3.1}$$

The capacitor then discharges through R_5, causing V_- to decrease. When V_- falls below $V_+ = 1.5$ V, the condition $V_+ > V_-$ is satisfied again, and the comparator output switches High. Thus,

$$V_{\text{sig}} = 5 \text{ V}, \qquad V_+ = \frac{3+5}{2} = 4.0 \text{ V}, \qquad V_- = 1.5 \text{ V} \tag{3.2}$$

Therefore, the circuit repeatedly alternates between the states shown in Eq. 3.1 and Eq. 3.2, producing an oscillating signal at the comparator output. The resulting oscillation is then passed to the 74HC14AP Schmitt trigger inverter. Since V_{sig} switches between 0 V and 5 V, these voltage levels are sufficiently below and above the lower and upper threshold voltages of the following Schmitt trigger inverter, respectively. Therefore, the oscillating signal generated at the comparator output can be reliably recognized as a digital input signal by the ATmega328P. Therefore, the ATmega328P can reliably identify the presence of a capacitor by detecting this oscillating signal. As will be described later, no oscillation is generated when a resistor or an open circuit is connected as the CUT.

3.3 Open-Circuit Detection

Second, the operation of the comparator when an open circuit is connected as the CUT is examined. The corresponding circuit is shown below.

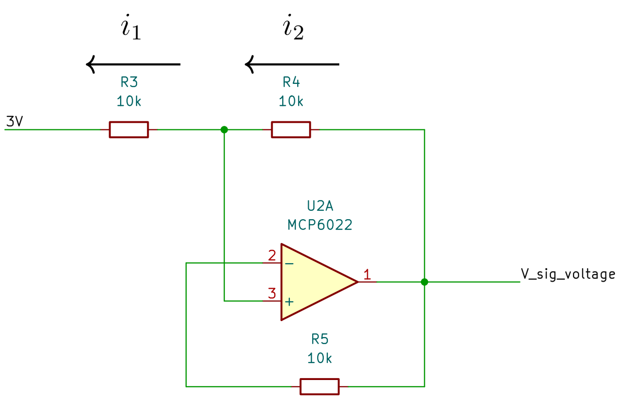

Figure 4: Comparator circuit when an open-circuit is connected as the CUT.

Since the MCP6022 is a rail-to-rail op-amp powered by a 5 V DC supply, the output voltage is assumed to remain within the supply rails: Since the MCP6022 is a rail-to-rail op-amp powered by a 5 V DC supply, the output voltage is assumed to remain within the supply rails:

$$0 < V_{\text{sig}} < 5 \text{ V}.$$

In addition, the feedback path from V_{sig} through R_5 to V_- forms a negative feedback loop. Therefore, the virtual short approximation is valid. Thus,

$$V_+ = V_-.$$

Applying Kirchhoff's current law at the non-inverting input node gives

$$\frac{V_+ - 3}{R_3} - \frac{V_{\text{sig}} - V_+}{R_4} = 0.$$

As $R_3 = R_4 = 10 \text{ k}\Omega$,

$$(V_+ - 3) - (V_{\text{sig}} - V_+) = 0.$$

Therefore,

$$2V_+ - 3 - V_{\text{sig}} = 0. \tag{3.3}$$

From the virtual short and no current flows over R_5,

$$V_+ = V_- = V_{\text{sig}}. \tag{3.4}$$

Using Eq. 3.4, Eq. 3.3 is

$$2V_{\text{sig}} - 3 - V_{\text{sig}} = 0.$$

Thus,

$$V_{\text{sig}} = 3 \text{ V}.$$

Therefore, in the open-circuit case,

$$V_+ = V_- = V_{\text{sig}} = 3 \text{ V}.$$

Therefore, at steady state, all nodes connected to the comparator are at 3 V. From Fig. 2, the comparator output voltage is above the upper threshold voltage of the following Schmitt trigger inverter by approximately 0.9 V. Therefore, when an open circuit is connected as the CUT, the ATmega328P recognizes a constant LOW signal from SIG_FREQUENCY without oscillation. As a result, the ATmega328P can distinguish an open circuit from a capacitor. As discussed in Section 3.4, the ATmega328P can distinguish an open circuit from a resistor by measuring SIG_VOLTAGE.

3.4 Resistor Detection

Finally, the operation of the comparator when a resistor is connected as the CUT is examined. The corresponding circuit is shown below.

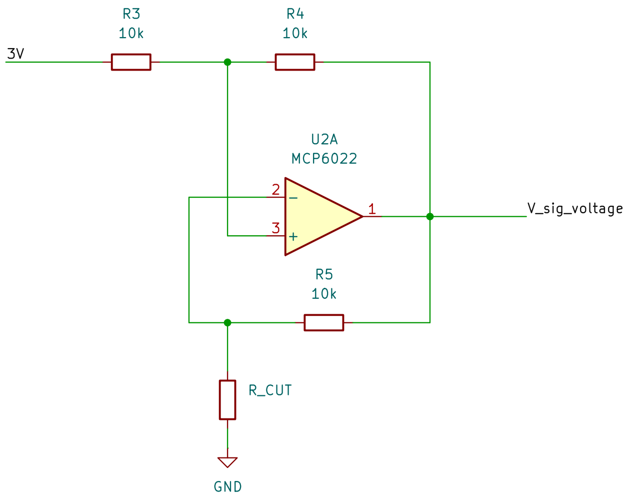

Figure 5: Comparator circuit when a resistor is connected as the CUT.

When a resistor is connected as the CUT, it is assumed that both the inverting input voltage V_- and the output voltage V_{sig} are initially 0 V. Therefore, immediately after power is applied, the voltage at the non-inverting input is

$$V_+ = \frac{3+0}{2} = 1.5\ \mathrm{V}.$$

Since $V_+ > V_-$ and a positive feedback path is formed from V_{sig} through R_4 to V_+, the output voltage and non-inverting input voltage become

$$V_{\mathrm{sig}} = 5\ \mathrm{V}, \qquad V_+ = \frac{3+5}{2} = 4.0\ \mathrm{V}.$$

At this point, the inverting input voltage V_- is determined by the voltage divider formed by R_5 and R_{CUT}. Therefore,

$$V_- = \frac{R_{\mathrm{CUT}}}{R_5 + R_{\mathrm{CUT}}} V_{\mathrm{sig}}.$$

For example, when $R_{\mathrm{CUT}} = 1\ \mathrm{k\Omega}$,

$$V_- = \frac{1\ \mathrm{k\Omega}}{10\ \mathrm{k\Omega} + 1\ \mathrm{k\Omega}} \times 5 = 0.45\ \mathrm{V}.$$

When $R_{\mathrm{CUT}} = 10\ \mathrm{k\Omega}$,

$$V_- = \frac{10\ \mathrm{k\Omega}}{10\ \mathrm{k\Omega} + 10\ \mathrm{k\Omega}} \times 5 = 2.5\ \mathrm{V}.$$

In either case, the relationship of $V_+ > V_-$ holds. Therefore, the output voltage remains at

$$V_{\mathrm{sig}} = 5\ \mathrm{V}.$$

Therefore, at steady state, the output voltage of the comparator is 5 V. From Fig. 2, the comparator output voltage is above the upper threshold voltage of the following Schmitt trigger inverter by approximately 2.09 V. Therefore, when a resistor is connected as the CUT, the ATmega328P recognizes a constant LOW signal from SIG_FREQUENCY without oscillation. As a result, the ATmega328P can distinguish a resistor from a capacitor. In addition, the ATmega328P can distinguish a resistor from an open circuit by measuring SIG_VOLTAGE.

3.5 LTspice Simulation Verification

In this section, the identification method discussed above is verified using LTspice simulations. The LTspice simulation schematic is shown in Fig. 6. The simulation was performed to check whether SIG_FREQUENCY oscillates only when a capacitor is connected, and whether SIG_VOLTAGE can distinguish between the open-circuit and resistor cases. Detailed simulation results are provided in Appendix A.

In the simulation, universal op-amp models and an SN74HC14 model were used instead of the actual NJU7043D, MCP6022, and TC74HC14AP devices. However, this simplification is acceptable because the simulation is intended to verify the logical operation of the circuit, not the exact device characteristics. The main concern is the threshold voltage of the Schmitt trigger inverter. The SN74HC14 model has an upper threshold voltage of approximately 2.77 V, while the upper threshold voltage of the TC74HC14AP is 2.91 V, as shown in Fig. 2. Since the minimum theoretical SIG_VOLTAGE is 3 V, both devices give the same logic result.

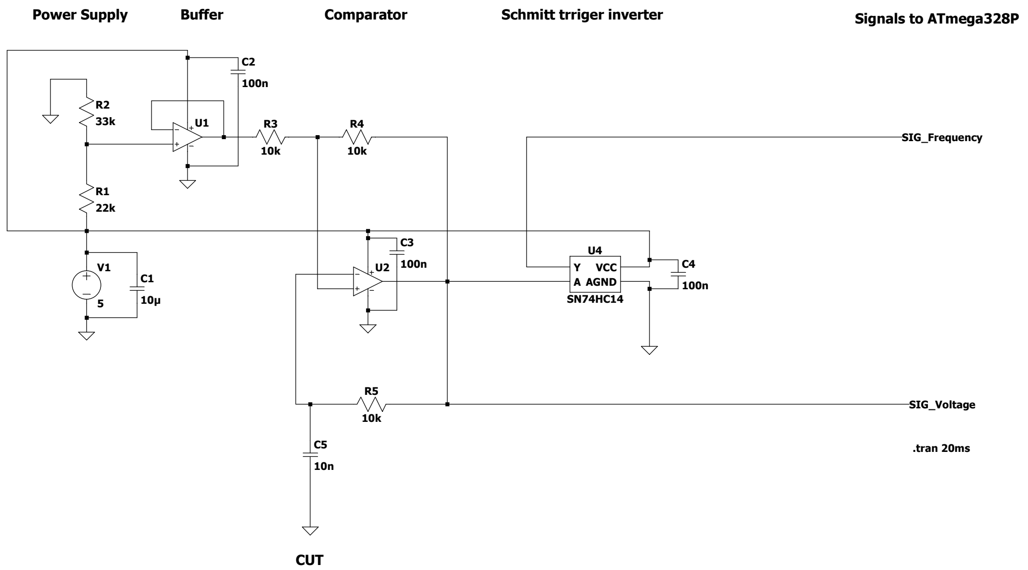

Figure 6: LTspice simulation circuit.

As shown in Appendix A.1, SIG_FREQUENCY oscillates only when a capacitor is connected as the CUT, while it remains at a Low voltage level when a resistor or an open circuit is connected. This is consistent with the theoretical operation discussed in Sections 3.2–3.4.

Also, as shown in Appendix A.2, SIG_VOLTAGE is approximately 3 V when an open circuit is connected, while it is approximately 5 V when a resistor is connected. This is consistent with the theoretical results discussed in Sections 3.3–3.4.

Based on these results, the theoretical discussions in Sections 3.2–3.4 are verified. Therefore, the identification method described in Section 3.1 is valid.

3.6 Capacitance Measurement

In this section, an equation for calculating the capacitance from the measured period is derived. Here, V_{T+} and V_{T-} denote the upper and lower threshold voltages for the capacitor charging and discharging operation, respectively. Also, V_H and V_L denote the High and Low output voltages of the MCP6022 as a part of the comparator, respectively.

Using Eq. B.6, the oscillation period is expressed as

$$T = t_{\text{charge}} + t_{\text{discharge}}.$$

For the charging interval,

$$t_{\text{charge}} = RC \ln\left(\frac{V_{T-} - V_H}{V_{T+} - V_H}\right).$$

For the discharging interval,

$$t_{\text{discharge}} = RC \ln\left(\frac{V_{T+} - V_L}{V_{T-} - V_L}\right).$$

Therefore,

$$T = RC\left[\ln\left(\frac{V_{T-} - V_H}{V_{T+} - V_H}\right) + \ln\left(\frac{V_{T+} - V_L}{V_{T-} - V_L}\right)\right].$$

Let the logarithmic term be a constant K. Hence,

$$T = RCK. \tag{3.5}$$

Thus, the capacitance is calculated as

$$C = \frac{T}{RK}. \tag{3.6}$$

The ATmega328P calculates the capacitance using this equation and the measured oscillation period. In the next section, the values of K and R are determined based on the measurement results to finalize the capacitance conversion equation.

4 Circuit-Level Measurement

In this section, the implemented circuit is first verified to confirm that it produces valid signals at SIG_FREQUENCY and SIG_VOLTAGE for identification, as described in Sections 4.1 and 4.2. Then, the constant K is determined based on the actual measurement results in Section 4.3. For reference, the raw measurement data are available in the following Excel files: "Section4.1_4.2_Measurement_Data.xlsx" and "Section4.3_Measurement_Data.xlsx".

4.1 Verification of Identification Using SIG_FREQUENCY

Table 1 summarizes the measurement results of SIG_FREQUENCY when a capacitor, an open circuit, or a resistor is connected as the CUT based on 15 times measurement. Capacitors of 1 nF, 4.7 nF, and 10 nF, and resistors of 1 kΩ, 4.7 kΩ, and 10 kΩ were tested as representative values. These values were selected to include both the lower and upper limits, as well as a middle value, in order to examine the overall behaviour of the circuit.

As shown in Table 1, oscillation was observed only when a capacitor was connected as the CUT. For the open-circuit and resistor cases, the Schmitt trigger inverter output remained at a Low voltage level. Therefore, the ATmega328P can identify the capacitor case using SIG_FREQUENCY.

Table 1: Measured output of the Schmitt trigger inverter for component identification.

CUT	SIG_FREQUENCY / Output State	Measured Output Voltage (V)
Capacitor	Oscillation observed	–
Open circuit	Low	0.013
Resistor, $R = 10$ kΩ	Low	-0.366
Resistor, $R = 4.7$ kΩ	Low	-0.045
Resistor, $R = 1$ kΩ	Low	-0.320

4.2 Verification of Identification Using SIG_VOLTAGE

Table 2 summarizes the measurement results of SIG_VOLTAGE when an open circuit or a resistor is connected as the CUT based on 15 times measurement. Resistors of 1 kΩ, 4.7 kΩ, and 10 kΩ were tested as representative values in order to examine the overall behaviour of the circuit..

Table 2: Measured SIG_VOLTAGE for open-circuit and resistor cases.

CUT	Measured SIG_VOLTAGE (V)
Open circuit	2.99
Resistor, $R = 10$ kΩ	4.69
Resistor, $R = 4.7$ kΩ	4.71
Resistor, $R = 1$ kΩ	4.66

As shown in Table 2, there is a voltage gap of more than approximately 1.6 V between the open-circuit and resistor cases. Therefore, by setting a threshold voltage between these two levels, the ATmega328P can distinguish between the resistor and open-circuit cases. As described in Section 5, a threshold voltage of 3.75 V was selected.

4.3 Determination of K

Table 3 summarizes the measured values required to determine the constant K. The measured values listed in Table 3 were obtained by averaging 15 measurements taken with capacitors of 1 nF, 2.2 nF, 4.7 nF, and 10 nF.

Table 3: Measured values used for determining the constant K.

Parameter	Description	Measured Value
V_{T+}	Upper threshold voltage	3.99 V
V_{T-}	Lower threshold voltage	1.46 V
V_H	High output voltage of the comparator	4.94 V
V_L	Low output voltage of the comparator	0.019 V

Substituting the values V_{T+}, V_{T-}, V_H, and V_L in Table 3 into the definition of K gives

$$K = \ln\left(\frac{V_{T-} - V_H}{V_{T+} - V_H}\right) + \ln\left(\frac{V_{T+} - V_L}{V_{T-} - V_L}\right) = \ln\left(\frac{1.46 - 4.94}{3.99 - 4.94}\right) + \ln\left(\frac{3.99 - 0.019}{1.46 - 0.019}\right) = 2.312.$$

The resistance value was determined as the average of 60 measurements, resulting in

$$R = 9.985\ \text{k}\Omega.$$

Therefore, from Eq. (3.5),

$$
\begin{aligned}
T &= RKC, \\
T &= (9.985)(2.312)C, \\
T &= 23.085\,C, \qquad (4.1)
\end{aligned}
$$

where R is expressed in kΩ, C in nF, and T in μs. Hence, the capacitance can be calculated as

$$C(\mathrm{nF}) = \frac{T(\mu\mathrm{s})}{23.085} \qquad (4.2)$$

The following table shows each capacitance threshold voltages based on 15 trials each.

Table 4: Measured threshold voltages for different capacitances.

Nominal Capacitance (nF)	Averaged Measured V_{T+} (V)	Averaged Measured V_{T-} (V)
1.0	4.01	1.44
2.2	3.99	1.46
4.7	3.98	1.47
10.0	3.98	1.48

As shown in Table 4, the measured threshold voltages vary slightly with capacitance. In particular, the difference between V_{T+} and V_{T-} decreases as the capacitance increases. This variation may be caused by a small delay in the comparator and the Schmitt trigger inverter. When the capacitance is smaller, the capacitor charges and discharges more quickly. Therefore, the trend may be observed here. In contrast, V_H and V_L remain approximately constant over the capacitance range from 1 nF to 10 nF. Since the constant K is determined using the average values of these parameters, the resulting capacitance conversion equation is expected to be most accurate near the middle of the measurement range. Consequently, larger errors are expected for 1 nF and 10 nF capacitors, while smaller errors are expected for 2.2 nF and 4.7 nF capacitors.

This tendency is expected from the averaging process used to determine K. Therefore, observing this tendency in the measured capacitance errors can be regarded as one indicator that the experiment was conducted properly and that the collected data are reasonably accurate. This point is discussed further in Section 7.

5 System-Level Implementation

In this section, the core circuit discussed above is integrated into a system by introducing an ATmega328P microcontroller, an OLED display, and a switch. Once the ATmega328P detects a LOW-level signal from the switch line, it processes the signals, `SIG_FREQUENCY`, `SIG_VOLTAGE` from the core circuit and displays the results on the OLED display. The detailed measurement algorithm for this process is provided in Appendix C.

5.1 Core Design Ideas of SIG_FREQUENCY Processing

To maximize measurement accuracy, Timer 1 of the ATmega328P and its Input Capture pin are used to determine frequencies. In the implemented circuit, a clock frequency of

16 MHz is used. Hence, the duration of one clock cycle (T_{clk}) is determined as follows:

$$T_{\mathrm{clk}} = \frac{1}{16\ \mathrm{MHz}} = 62.5\ \mathrm{ns}. \tag{5.1}$$

When $C = 1$ nF, the period is the shortest. From Eq. (4.1), the minimum period ($T_{1\mathrm{nF}}$) is determined as follows:

$$T_{1\mathrm{nF}} = 23.085 \times 1 = 23.085\ \mu\mathrm{s}. \tag{5.2}$$

Therefore, considering the clock resolution of 62.5 ns, the minimum number of captured clock counts (N_{min}) within one period can be calculated as follows:

$$N_{\mathrm{min}} = \frac{23.085\ \mu\mathrm{s}}{62.5\ \mathrm{ns}} \approx 369.36. \tag{5.3}$$

Because the timer counts in steps of 1, the measurement error is limited to ± 1 clock count. The worst-case error (E_{max}) is obtained as follows:

$$E_{\mathrm{max}} = \frac{1}{N_{\mathrm{min}}} \approx \frac{1}{369.36} \approx 0.27\%. \tag{5.4}$$

As a result, it is confirmed that the system can achieve a high accuracy of over 99.7%.

Additionally, to reduce outliers such as those affected by noise, 200 samples are collected and sorted. Then, the 20 samples in the middle are averaged to determine the final frequency measured by ATmega328P.

5.2 Core Design Ideas of SIG_VOLTAGE Processing

In the same manner as the sampling method in Section 5.1, 200 samples are collected and sorted. The middle 20 samples are then averaged to determine the final value of SIG_VOLTAGE measured by the ATmega328P.

In addition, 3.75 V is set as the threshold voltage to distinguish between an open circuit and a resistor.

6 System-Level Measurement Results

6.1 Identification Results

The following table shows the experimental results of the implemented system based on 5 trials for each case. The results demonstrate that the system works as expected within the ranges of 1 $k\Omega$ to 10 $k\Omega$ and 1 nF to 10 nF. Furthermore, when the resistance range was expanded from 220 Ω to 47 $k\Omega$, the system successfully and accurately identified the components in all cases.

For reference, the detailed experimental data is available in `Section_6.1_Measurement_Data.xlsx`.

6.2 Results of measured Values

This section presents the measured periods, capacitances.

Table 5: System-level identification results.

CUT	Trials	Result
Open circuit	5	OPEN (5/5)
Resistor, $R = 1$ kΩ	5	RESISTOR (5/5)
Resistor, $R = 2.2$ kΩ	5	RESISTOR (5/5)
Resistor, $R = 4.7$ kΩ	5	RESISTOR (5/5)
Resistor, $R = 10$ kΩ	5	RESISTOR (5/5)
Capacitor, $C = 1$ nF	5	CAPACITOR (5/5)
Capacitor, $C = 2.2$ nF	5	CAPACITOR (5/5)
Capacitor, $C = 4.7$ nF	5	CAPACITOR (5/5)
Capacitor, $C = 10$ nF	5	CAPACITOR (5/5)

The following table shows the periods measured by an oscilloscope, the values displayed on the OLED, and their corresponding errors. Both the measured and displayed periods were obtained from five trials for each capacitance. Finally, the period error read by the ATmega328P ranges from 0.04 % to 0.4 %. As shown in Equation (5.4), the maximum theoretical error due to quantization is about 0.27 %. The errors for 1.0 nF, 4.7 nF, and 10 nF are smaller than this limit. However, the error for 2.2 nF is larger than the value.

Possible causes for this discrepancy include measurement inaccuracies and non-ideal characteristics of the components. For instance, the circuit may have been subjected to mechanical vibrations during switching, or functioned under ambient electrical noise such as my body temperature. Additionally, the effects of parasitic capacitance could contribute to the error.

However, overall, the periods read by the ATmega328P can be considered fairly accurate. The utilization of the Input Capture likely played a crucial role in achieving this high precision.

Table 6: Period Measurement Results

Capacitance (nF)	Measured Period (μs)	Displayed Period (μs)	Error (%)
1.0	23.92	23.94	0.08
2.2	47.14	47.33	0.40
4.7	102.64	102.57	0.07
10.0	258.73	258.83	0.04

The following table shows the capacitances measured 15 times by a digital multimeter, the values displayed on the OLED based on five trials, and their corresponding errors. Finally, the total system error ranges from 0.97 % to 3.5 % across the evaluated capacitances (1 nF, 2.2 nF, 4.7 nF, and 10 nF).

Based on the results presented above, the average processing effect discussed in Section 4.3 cannot be clearly observed. It is therefore likely that other factors affected the measurement results. A more detailed discussion is provided in Section 7, where the breakdown of the observed error is examined in detail.

For reference, the detailed experimental data is available in `Section_6.2_Measurement_Data.xlsx`.

Table 7: Capacitance Measurement Results

Nominal Capacitance (nF)	Measured Capacitance (nF)	Displayed Capacitance (nF)	Error (%)
1.0	1.02	1.04	1.96
2.2	2.07	2.05	0.97
4.7	4.61	4.45	3.50
10.0	10.88	11.21	3.00

7 Error Analysis

This section addresses the errors presented in Table 7 by evaluating the primary error sources that have emerged through the process thus far. Next, it is investigated whether the residual errors can be explained by small factors such as some non-ideal factors. The diagram below illustrates the primary error sources mentioned earlier.

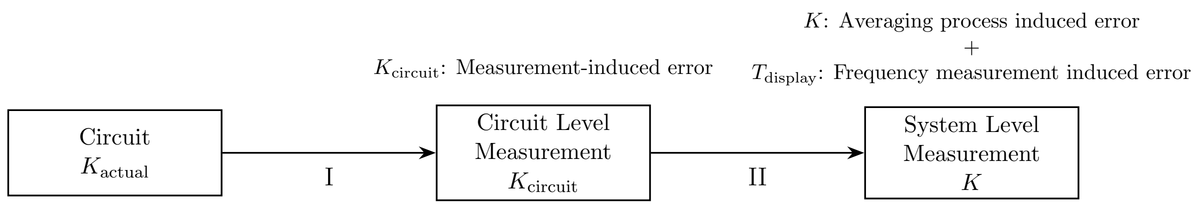

Figure 7: Measurement processes and primary error sources.

Where each coefficient and parameter is defined as follows:

- K_{actual}: The true coefficient K.
- K_{circuit}: The coefficient derived from calculations based on the necessary node voltages measured during the circuit-level measurement phase.
- K: The average value of the coefficients derived during the circuit-level measurement process across the different capacitance values (1 nF, 2.2 nF, 4.7 nF, and 10 nF).
- T_{display}: The period derived based on the frequency measured by the ATmega328P.

Process I corresponds to Section 5, and Process II corresponds to Section 5. Process I is a potential source of error, as measurement accuracy constraints can lead to errors in the derived coefficients for each capacitance value. Process II serves as another potential source of error. This is because the averaging process is applied, and the frequency acquired by the ATmega328P is generally expected to contain errors relative to the true frequency, which can potentially introduce errors into its reciprocal, the period T_{display}. From here, we calculate the error derived from Process I and II.

The total error $E_{\text{display-10nF}}$ of the displayed capacitance $C_{\text{display-10nF}}$ relative to the actual capacitance $C_{\text{actual-10nF}}$ is evaluated.

$$
\begin{aligned}
E_{\text{display-10nF}} &= \frac{C_{\text{display-10nF}} - C_{\text{actual-10nF}}}{C_{\text{actual-10nF}}} \times 100 \\
&= \frac{\frac{T_{\text{display-10nF}}}{R_{\text{actual}} \cdot K} - \frac{T_{\text{actual-10nF}}}{R_{\text{actual}} \cdot K_{\text{actual-10nF}}}}{\frac{T_{\text{actual-10nF}}}{R_{\text{actual}} \cdot K_{\text{actual-10nF}}}} \times 100 \\
&= \left(\frac{K_{\text{actual-10nF}}}{K} \times \frac{T_{\text{display-10nF}}}{T_{\text{actual-10nF}}} - 1 \right) \times 100 \qquad (7.1)
\end{aligned}
$$

From the measured data,

$$\frac{T_{\text{display-10nF}}}{T_{\text{actual-10nF}}} = \frac{258.83}{258.73} \tag{7.2}$$

The equation for the error $E_{\text{circuit-10nF}}$ of $C_{\text{circuit-10nF}}$ relative to $C_{\text{actual-10nF}}$:

$$E_{\text{circuit-10nF}} = \left(\frac{K_{\text{actual-10nF}}}{K_{\text{circuit-10nF}}} \times \frac{T_{\text{circuit-10nF}}}{T_{\text{actual-10nF}}} - 1 \right) \times 100 \tag{7.3}$$

From the measured data,

$$E_{\text{circuit-10nF}} = 0 \tag{7.4}$$

From the experimental data,

$$\frac{T_{\text{circuit-10nF}}}{T_{\text{actual-10nF}}} = \frac{247}{259} \tag{7.5}$$

From (7.3), (7.4), (7.5),

$$\frac{K_{\text{actual-10nF}}}{K_{\text{circuit-10nF}}} = \frac{259}{247} \qquad \therefore K_{\text{actual-10nF}} = \frac{259}{247} K_{\text{circuit-10nF}} \tag{7.6}$$

Based on the experimental data,

$$\frac{K_{\text{circuit-10nF}}}{K} = \frac{2.273}{2.312} \qquad \therefore K_{\text{circuit-10nF}} = \frac{2.273}{2.312} K \tag{7.7}$$

From (7.6), (7.7),

$$\frac{K_{\text{actual-10nF}}}{K} = \frac{259}{247} \times \frac{2.273}{2.312} \approx 1.0309 \tag{7.8}$$

From (7.1), (7.2), and (7.8),

$$
\begin{aligned}
E_{\text{display-10nF}} &= \left(1.0309 \times \frac{258.83}{258.73} - 1 \right) \times 100 \qquad (7.9) \\
&\approx 3.13\% \qquad (7.10)
\end{aligned}
$$

From the measured data, $C_{\text{display-10nF}} = 11.21$ nF and $C_{\text{actual-10nF}} = 10.88$ nF, where an error of +3% is observed. Therefore, the residual error $\Delta E_{\text{display-10nF}}$ is:

$$\Delta E_{\text{display-10nF}} = 3\% - 3.13\% = -0.13\% \tag{7.11}$$

In the same fashion, the residual errors for the 1.0 nF, 2.2 nF, and 4.7 nF capacitors are calculated. A summary of the results is presented in Table 8.

Table 8: Summary of final, total, and residual errors for each capacitance.

Capacitance (nF)	Final error (%)	Total error (I, II) (%)	Residual error (%)
1.0	+1.96	+2.18	-0.22
2.2	-0.97	-0.99	+0.02
4.7	-3.50	-2.78	-0.72
10.0	+3.00	+3.13	-0.13

The remaining errors are all below 1% and do not exhibit any systematic trend with capacitance. Therefore, the residual errors are likely to be caused by random uncertainties such as electrical noise, connection instability, parasitic effects, and other measurement uncertainties.

Table 9: Breakdown of the total error into Process I and Process II contributions.

Capacitance (nF)	Process I error (%)	Process II error (%)	Total error (I, II) (%)
1.0	+0.980	+1.199	+2.18
2.2	+0.483	-1.476	-0.99
4.7	0.000	-2.776	-2.78
10.0	0.000	+3.130	+3.13

The following table presents a breakdown of the total error into Process I and Process II contributions. In this table, the errors for Process I and Process II are presented to three decimal places, whereas the total error (I, II) is presented to two decimal places. The Process I error is calculated using

$$E_{\mathrm{I}} = \frac{C_{\mathrm{circuit}} - C_{\mathrm{actual}}}{C_{\mathrm{actual}}} \times 100,$$

where C_{circuit} is the capacitance calculated from the conversion equation based on K_{circuit}, and C_{actual} is the capacitance measured using a digital multimeter. The Process II error is calculated as the difference between the total error and the Process I error.

The Process I error originates from the measurements required to determine C_{actual} and to calculate C_{circuit}. For the 1.0 nF and 2.2 nF capacitors, the derived capacitance values differed slightly from the actual capacitance values, while for the 4.7 nF and 10 nF capacitors, the derived and actual capacitance values were identical within the displayed precision. This suggests that the measurements required to obtain C_{circuit} were sufficiently accurate for the 4.7 nF and 10 nF capacitors. In contrast, the discrepancies observed for the 1.0 nF and 2.2 nF capacitors indicate the presence of measurement-related errors.

The Process II error originates from the averaging process and the frequency measurement performed by the ATmega328P. Larger errors were expected for the 1.0 nF and 10 nF capacitors, while smaller errors were expected for the 2.2 nF and 4.7 nF capacitors. However, this trend was not clearly observed. This suggests that additional factors may have contributed to the error after Process I.

As a result, this error analysis suggests that the residual error can be attributed to measurement uncertainties and that most of the observed errors originate from Process I and Process II.

For reference, the detailed experimental data are available in `Section7_Measurement_Data.xlsx`.

8 Conclusion

In this project, a system capable of identifying resistors, capacitors, and open circuits was successfully designed and implemented. The system was able to measure capacitance in the range of 1 nF to 10 nF with an accuracy of 0.97% to 3.5%.

As discussed in Section 7, most of the error was attributed to the process after the ATmega328P received the signals from the main circuit. Although the expected averaging effect was not clearly observed, this process is considered to be the main limitation of the implemented system. Therefore, the capacitance measurement accuracy is expected to decrease outside the tested range of 1 nF to 10 nF.

One possible improvement is to measure the node voltages required to determine the conversion coefficient K, such as V_{T+}, V_{T-}, V_H, and V_L, each time a capacitor is connected as the CUT. By calculating K under the actual circuit conditions each time, the averaging process and the preliminary determination of K_{circuit} would no longer be required. Therefore, the corresponding error sources in Process II and Process I would be eliminated. While the frequency measurement error of the ATmega328P would still remain, additional uncertainty would be newly introduced by the required node voltage measurements. This approach would improve the capacitance measurement accuracy and extend the measurable capacitance range.

A LTspice Simulation Results

A.1 SIG_FREQUENCY Results

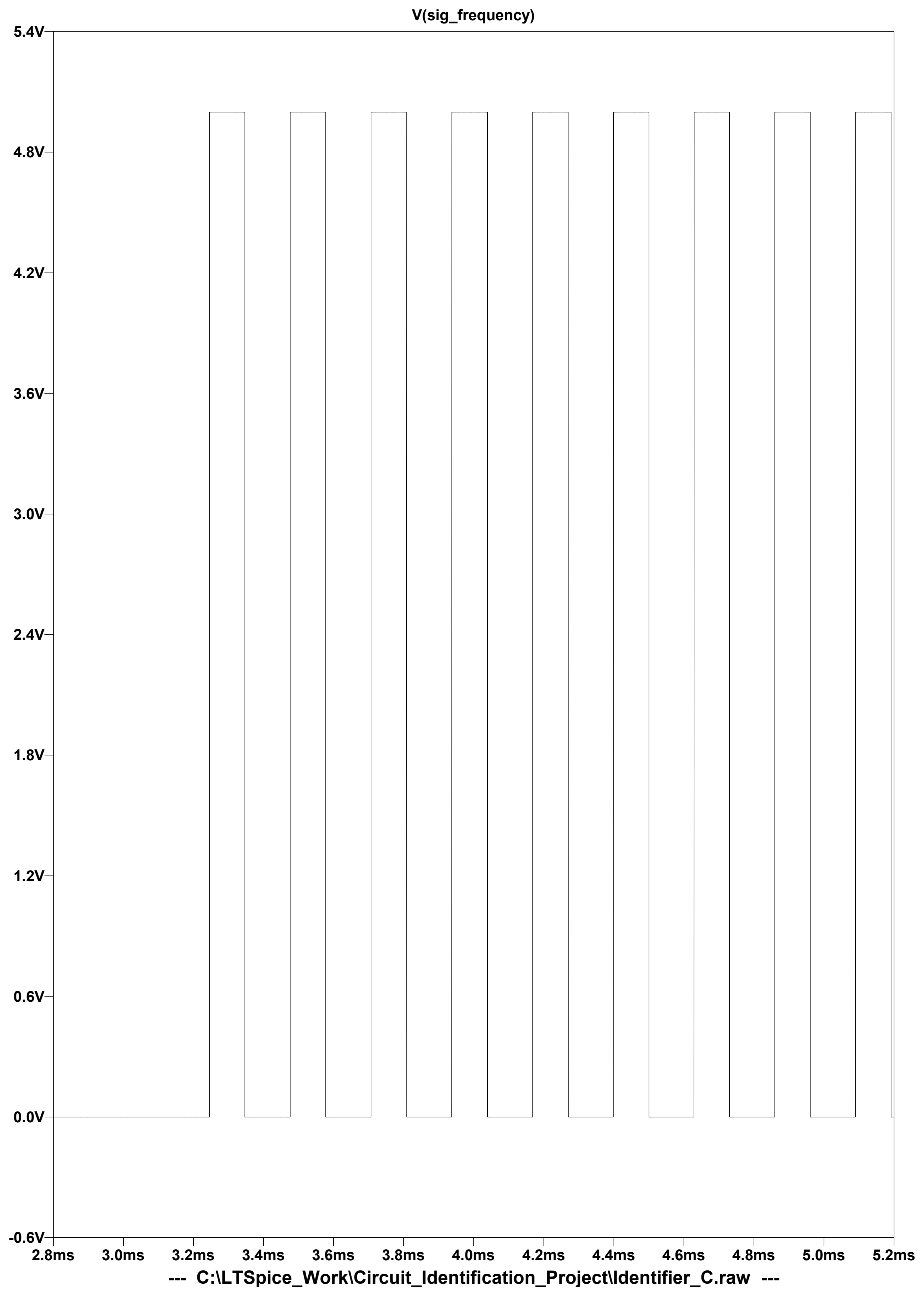

Figure 8: LTspice simulation result of SIG_FREQUENCY for the capacitor case.

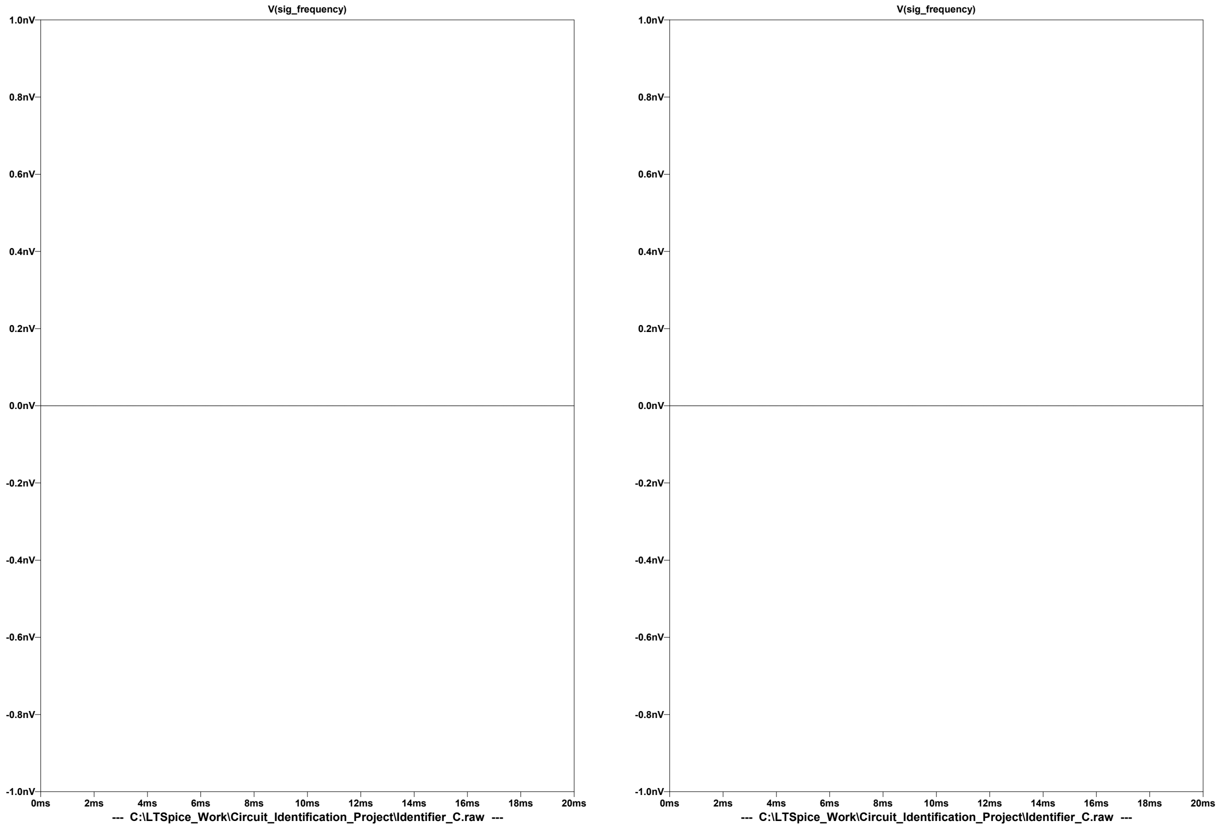

Figure 9: LTspice simulation results of SIG_FREQUENCY for open-circuit and resistor cases.

A.2 SIG_VOLTAGE Results

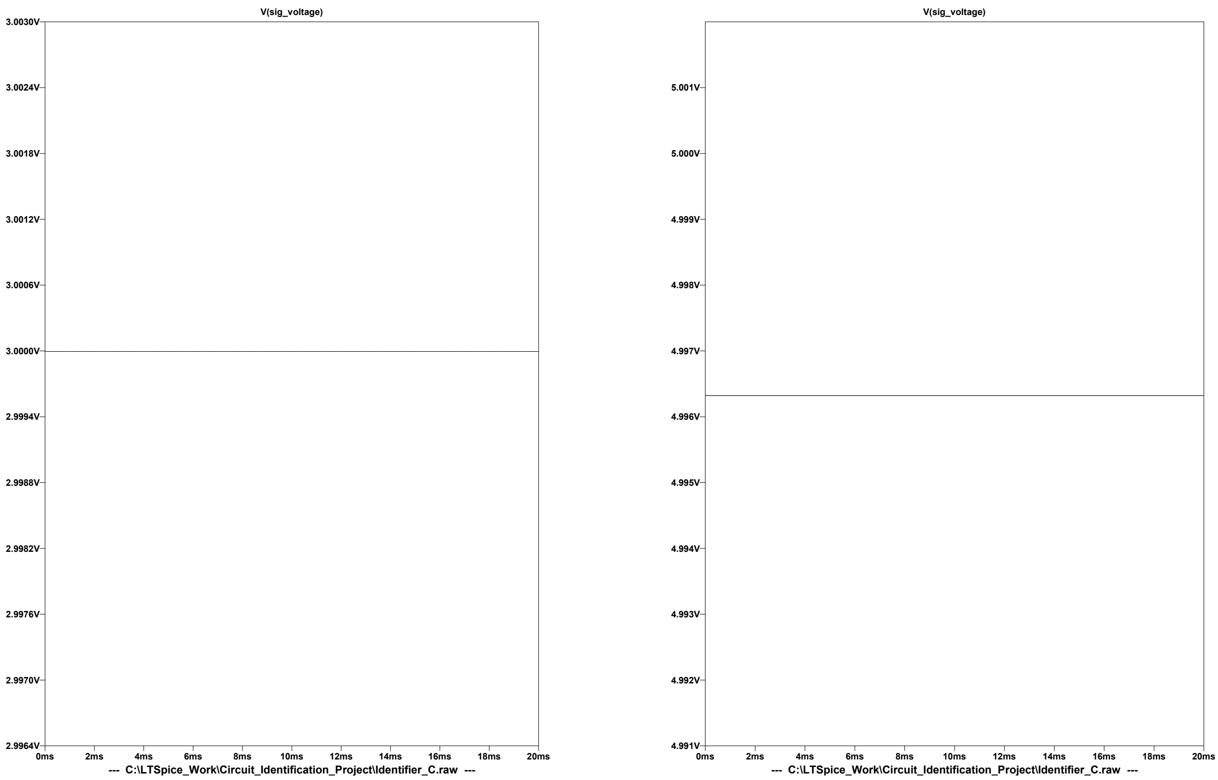

Figure 10: LTspice simulation results of SIG_VOLTAGE for open-circuit and resistor cases.

B Derivation of RC Equations

B.1 Derivation of Capacitor Voltage Equation

The capacitor charging and discharging equations used in Section 3.6 are derived in this section. The RC circuit model used for the derivation is shown in Fig. 11. Applying

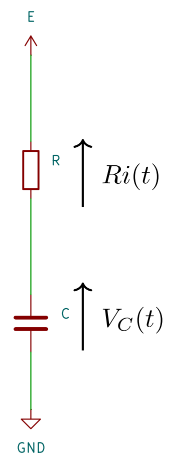

Figure 11: RC circuit model.

Kirchhoff's voltage law to the RC circuit in Fig. 11 gives

$$E = Ri(t) + V_C(t). \tag{B.1}$$

Since the current through the capacitor is given by

$$i(t) = C\frac{dV_C(t)}{dt},$$

Eq. B.1 becomes

$$E = RC\frac{dV_C(t)}{dt} + V_C(t).$$

Rearranging this equation gives

$$\frac{dV_C(t)}{dt} + \frac{1}{RC}V_C(t) = \frac{E}{RC}.$$

Using the general solution of a first-order linear differential equation with constant coefficients, the capacitor voltage can be written as

$$V_C(t) = E + Ae^{-t/RC}, \tag{B.2}$$

where A is a constant determined by the initial condition.

Let the initial value of the capacitor voltage be

$$V_C(0) = V_1. \tag{B.3}$$

Substituting Eq. B.3 into Eq. B.2 gives

$$V_1 = E + A.$$

Therefore,

$$A = V_1 - E.$$

Thus, the capacitor voltage is expressed as

$$V_C(t) = E + (V_1 - E)e^{-t/RC}. \tag{B.4}$$

B.2 Derivation of RC Charging and Discharging Time Equation

Using Eq. B.4, the time required for the capacitor voltage to change from V_1 to V_2 under the supply voltage E is derived. When $V_C(t) = V_2$,

$$V_2 = E + (V_1 - E)e^{-t/RC}. \tag{B.5}$$

Rearranging Eq. B.5 gives

$$V_2 - E = (V_1 - E)e^{-t/RC}.$$

Therefore,

$$e^{-t/RC} = \frac{V_2 - E}{V_1 - E}.$$

When charging, both V_1 and V_2 are smaller than E. When discharging, both V_1 and V_2 are larger than E. Therefore, in both cases, the following condition holds.

$$\frac{V_2 - E}{V_1 - E} > 0.$$

Hence, taking the natural logarithm of both sides gives

$$-\frac{t}{RC} = \ln\left(\frac{V_2 - E}{V_1 - E}\right).$$

Thus, the time required for the capacitor voltage to change from V_1 to V_2 under the supply voltage E is

$$t = RC \ln\left(\frac{V_1 - E}{V_2 - E}\right). \tag{B.6}$$

C Measurement Algorithm

The algorithm implemented in the ATmega328P is summarized as follows.

1. After power is applied, the system waits for 1 s to avoid the power-on transient.
2. The OLED displays `WAITING`.
3. A 100 ms threshold is used to determine a valid switch `ON` state for stability.
4. When the ATmega328P detects the switch is turned `ON`, the OLED displays `MEASURING`.
5. SIG_FREQUENCY is observed for up to 500 ms.
6. If 200 valid period samples are obtained within the valid period range, the connected component is identified as a capacitor.
   - 6.1 For the capacitor case, the 200 period samples are sorted, and the average of the central 20 samples, corresponding to indices 90–109, is calculated.
   - 6.2 The capacitance is calculated using
   $$C(\mathrm{nF}) = \frac{T(\mu\mathrm{s})}{23.085}.$$
   - 6.3 The OLED displays the calculated capacitance C in nF and the measured period T in μs.
7. If 200 valid period samples are not obtained, 200 samples of SIG_VOLTAGE are acquired.
   - 7.1 The 200 voltage samples are sorted, and the average of the central 20 samples, corresponding to indices 90–109, is calculated.
   - 7.2 A threshold voltage of 3.75 V is used to distinguish between the resistor and open-circuit cases.
   - 7.3 The OLED displays either `RESISTOR` or `OPEN`, together with the measured voltage.
8. Once the ATmega328P detects a stable switch `OFF` state for 100 ms, the system returns to the `WAITING` state.

D Implemented Circuit Photograph

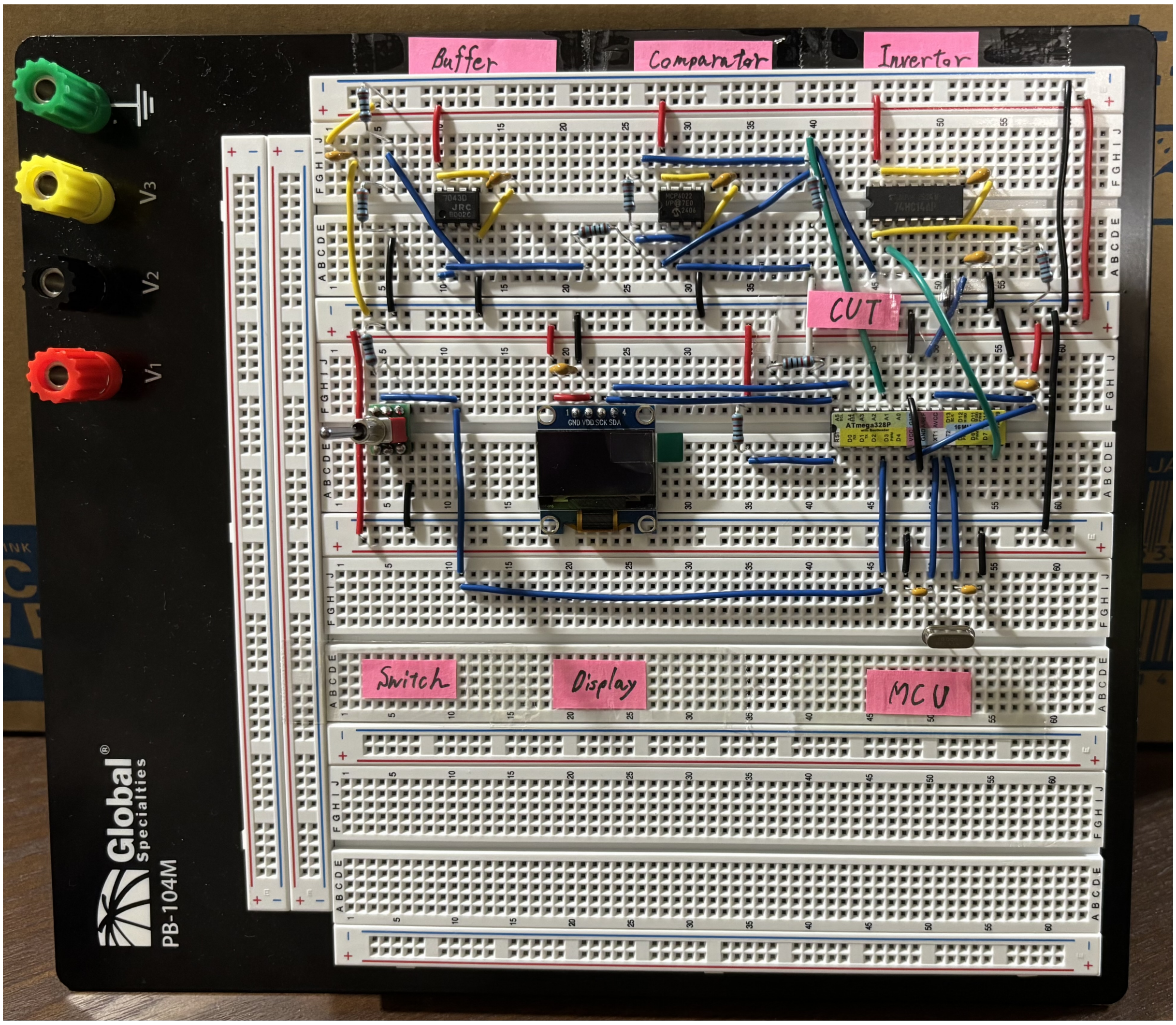

Figure 12: Photograph of the implemented circuit in this project.

The detailed colour coding of the wiring used in the prototype is summarized in Table 10.

Table 10: Colour coding of wiring used in the prototype

Colour	Code	Purpose
Red		+5 V power distribution.
Black		Ground (GND) connections.
Yellow		Wiring associated with noise suppression circuits.
Green		Signal lines carrying measurement information, such as `SIG_FREQUENCY`, `SIG_VOLTAGE`.
White		Connections to the Component Under Test (CUT).
Blue		General circuit interconnections.